MILITARY SPECIFICATION

WIRING HARNESS, BRANCHED, FOR M4 CONTROL UNIT

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of wiring harness entitled Wiring Harness, Branched which is used in the Sergeant warhead section.

SPECIFICATIONS

MILITARY

MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for.

DRAWINGS

ORDNANCE CORPS

8830468	- Wiring Harness, Branched.
EL-8854163	- Equipment Lists.

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PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The harness shall comply with all requirements specified on the drawing (dwg) 8830468 , with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SMUPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

3.5 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

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3.6 Continuity and resistance. The continuity and resistance between the following terminals of the wiring harness shall not exceed 0.1 ohms when tested as specified in 4.4.1.

From Terminal	To Terminal
2J1-B	2J3-v
2J1-C	2J1-D
2J1-E	2J4-p
2J1-G	2J4-E
2J1-G	2J4-r
2J1-G	2P2-L
2J1-H	2P3-e
2J1-K	2P3-t
2J1-L	2P3-Z
2J1-M	2P3-S
2J1-R	2J4-K
2J1-U	2P2-p
2J1-V	2P2-q
2J1-e	2P2-k
2J1-j	2J3-f
2J1-j	2J4-n
2J1-k	2P2-c
2J1-m	2P3-p
2J1-n	2P2-g
2J1-p	2P3-r
2J1-q	2P3-m
2J1-q	Case
2J1-r	2P2-i
2J1-s	2P3-m
2J2-Center	2P3-n
2J3-A	2J5-g
2J3-B	2P1-B
2J3-C	2P1-W
2J3-D	2P1-Y
2J3-D	2P3-y
2J3-E	2J7-F
2J3-F	2J7-U
2J3-G	2J5-b
2J3-H	2J7-S
2J3-J	2P2-G
2J3-K	2P2-R
2J3-L	2P1-E
2J3-L	2P1-H
2J3-L	2P1-K
2J3-L	2P2-F

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From Terminal	To Terminal
2J3-M	2P2-S
2J2-N	2P3-AA
2J3-R	2P4-F
2J3-T	2P2-V
2J3-U	2J7-D
2J3-U	2P3-L
2J3-V	2P2-W
2J3-W	2P2-X
2J3-Y	2J5-F
2J3-Y	2J5-Y
2J3-Z	Case
2J3-a	2P3-E
2J3-$\overline{b}$	2P2-b
2J3-$\overline{c}$	2P2-$\overline{h}$
2J3-$\overline{d}$	2P3-$\overline{Y}$
2J3-$\overline{e}$	2P5-M
2J3-$\overline{g}$	2J6-e
2J3-$\overline{h}$	2P5-$\overline{F}$
2J3-$\overline{i}$	2P3-a
2J3-$\overline{j}$	2P5-$\overline{D}$
2J3-$\overline{m}$	2P3-M
2J3-$\overline{n}$	2P3-b
2J3-$\overline{q}$	2P3-$\overline{BB}$
2J3-$\overline{q}$	2P1-b
2J3-$\overline{r}$	2P5-$\overline{C}$
2J3-s	2P3-d
2J3-$\overline{t}$	2P3-$\overline{D}$
2J3-$\overline{u}$	2P3-G
2J3-$\overline{w}$	2P3-R
2J3-$\overline{x}$	2P2-j
2J3-$\overline{y}$	2P2-$\overline{m}$
2J3-$\overline{z}$	2P2-$\overline{d}$
2J3-$\overline{AA}$	2P3-$\overline{T}$
2J3-BB	2P4-T
2J3-CC	2P3-K
2J3-DD	2P3-W
2J3-EE	2P3-j
2J3-FF	2P3-$\overline{U}$
2J3-GG	2P3-X
2J3-HH	2P3-c
2J4-A	2J5-$\overline{S}$
2J4-B	2J5-W
2J4-C	2J1-D
2J4-D	2P1-F

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From Terminal	To Terminal
2J4-F	2P1-P
2J4-G	2J5-G
2J4-G	2J6-G
2J4-H	2J5-V
2J4-L	2J5-H
2J3-S	2P3-u
2J3-X	2P3-$\overline{v}$
2J3-k	2P3-$\overline{w}$
2J4-$\overline{J}$	2P1-$\overline{V}$
2J4-M	2J6-V
2J4-N	2J6-W
2J4-R	2P1-Z
2J4-S	2J7-L
2J4-T	2J7-N
2J4-U	2J7-Z
2J4-V	2J7-V
2J4-W	2P1-M
2J4-X	2J7-P
2J4-Y	2J7-K
2J4-Z	2J7-M
2J4-a	2P1-e
2J4-$\overline{b}$	2J7-$\overline{b}$
2J4-$\overline{c}$	2J7-$\overline{c}$
2J4-$\overline{d}$	2P1-$\overline{L}$
2J4-$\overline{e}$	2P1-$\overline{j}$
2J4-$\overline{f}$	2P4-$\overline{L}$
2J4-$\overline{I}$	2P4-E
2J4-$\overline{j}$	2P2-P
2J4-$\overline{m}$	2J5-N
2J4-$\overline{m}$	2J6-N
2J5-$\overline{B}$	2P3-V
2J5-C	2J5-D
2J5-C	2P3-H
2J5-E	2J5-a
2J5-H	2J6-$\overline{H}$
2J5-J	2J6-J
2J5-K	2J6-K
2J5-L	2J6-L
2J5-R	2J6-S
2J5-T	2J5-U
2J5-T	2P3-F
2J5-X	2P3-A
2J5-Z	2J6-F
2J5-Z	2J6-Y

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From Terminal	To Terminal
2J5-c	2J6-b
2J5-d	2P3-z
2J5-e	2J3-P
2J5-f	2J6-g
2J6-C	2J6-D
2J6-C	2P3-C
2J6-E	2J6-a
2J6-R	2J7-A
2J6-T	2J6-U
2J6-T	2P3-B
2J6-X	2P3-k
2J6-Z	2P2-T
2J6-c	2P3-N
2J6-d	2P2-t
2J6-f	2P1-C
2J7-B	2P3-h
2J7-C	2P3-g
2J6-B	2P3-x
2J7-E	2P3-J
2J7-G	2P2-C
2J7-H	2P2-N
2J7-J	2P2-M
2J7-R	2P1-A
2J7-T	2P3-i
2J7-W	2P2-B
2J8-Center	2P3-q
2P1-N	2P2-A
2P1-S	2P4-D
2P1-T	2P2-K
2P1-a	2P2-r
2P1-c	2P2-s
2P1-f	2P2-E
2P1-g	2P5-H
2P2-D	2P5-G
2P2-H	2P4-S
2P2-J	2P5-B
2P2-Y	2P4-A
2P2-Z	2P4-C
2P2-a	2P5-K
2P2-e	2P5-J
2P2-f	2P4-U
2P2-n	2P5-L
2P3-P	2J3-p
2P3-s	2P5-E

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From Terminal	To Terminal
2P2-f	2P4-P
2P4-B	2P5-A
2P4-N	2P4-R
2P3-f	2P4-M

3.6.1 Circuit resistance. The circuit resistance of the wiring harness shall be as tabulated below and tested as specified in 4.4.1.1.

From Terminal	To Terminal	Resistance (Ohms)	Test Code Nos.
2J1-P	2J4-L	6960 to 7040	031001
2J1-S	2J4-L	4972 to 5028	031002
2P1-g	2P3-i	295 to 325	031003
2J4-L	2J5-J	1996 to 2004	031004
2J4-L	2J5-K	3992 to 4008	031005
2J4-L	2J5-L	5988 to 6012	031006
2P4-A	2P4-J	1900 to 2100	031007

3.7 Diode characteristics. Forward voltage drop across the diode and the reverse voltage drop across a 150 ± 1.5 ohm or 100 ± 1% kilohm resistor as applicable shall be as specified when the relays are energized with terminals 2P5-C and 2P5-D shorted to terminals 2P5-J and 2P5-L respectively. The testing shall be as specified in 4.4.2.

Positive Lead To Terminal	Negative Lead To Terminal	Applied Forward Current	Max Forward Volt Drop	Test Code Nos.
2J3-r	2J3-d	10 ± 0.5A	2.25	040001
2J3-j	2J3-d	10 ± 0.5A	2.25	040002
2P4-K	2J3-p	500 ± 25ma	1.2	040003
2P4-J	2J3-p	500 ± 25ma	1.2	040004

Positive Lead To Terminal	Negative Lead To Terminal	Applied Reverse Voltage	Max Volt Drop Across Series Res.	Test Code Nos.
2J3-d	2J3-r	100 ± 5.0	1.5	040005
2J3-d	2J3-j	100 ± 5.0	1.5	040006
2J3-p	2P4-K	200 ± 10.0	0.3	040007
2J3-p	2P4-J	200 ± 10.0	0.3	040008

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3.8 Insulation resistance. The insulation resistance shall be greater than 20 megohms between each of the following terminals and all other terminals listed below and ground when tested as specified in 4.4.3.

Following humidity testing, insulation resistance between each of the listed terminals and all other terminals listed shall be greater than 2 megohms. Testing shall be as specified in 4.4.3.

Receptacle 2J1 - B, C, E, G, H, K, L, M, P, R, U, V, e, j, k, m, n, p, r

Receptacle 2J3 - A, B, C, E, F, G, H, J, K, L, M, S, X, N, R, T, V, W, Y, a, b, c, d, e, g, h, k, i, j, m, n, p, q, r, s, t, u, w, x, y, z, AA, BB, CC, DD, EE, FF, GG, HH

Receptacle 2J4 - A, B, C, D, F, G, H, J, M, N, R, S, T, U, V, W, X, Y, Z, a, b, c, d, e, j, m

Receptacle 2J5 - B, D, E, R, T, X, Z, c, d, e, f

Receptacle 2J6 - B, D, E, R, U, X, Z, c, f, d

Receptacle 2J7 - B, C, D, E, G, H, J, R, T, W

Plug 2P1 - N, S, T, Y, a, c, f

Plug 2P2 - D, H, J, Z, a

Plug 2P3 - f, n, q, s

Plug 2P4 - B, E, L, N, U

Ground - This shall be test fixture ground.

Unused Pins

Receptacle 2J1 - A, F, J, N, T, W, X, Y, Z, a, b, c, d, f, g, h, i, t

Receptacle 2J4 - P, g, k, q, h

Receptacle 2J5 - A, M, P

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Receptacle 2J6 - A, M, P

Receptacle 2J7 - X, Y, a

Plug 2P1 - G, J, R, U, X, d, h

Plug 2P3 - CC, DD, EE, FF, GG, HH, JJ, KK, LL, MM, NN, PP

Plug 2P4 - V, G, H

Plug 2P2 - U

Unused terminals in any one connector shall be connected together and tested as one terminal.

Connector shells shall be at ground potential.

3.9 Environmental testing. The harness shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The harness shall also comply with 4.3.3.1 before and after each environmental test. Testing shall be as specified in 4.4.4 Tests shall be conducted in the order specified.

3.9.1 High temperature. The wiring harness shall withstand exposure to a temperature of 160 ± 5° Fahrenheit (F) without damage and while at this temperature shall be capable of meeting the requirements of 3.7 and 3.8. The wiring harness shall meet the requirements of 3.6 and 3.6.1 after completion of high temperature test at 80 ± 20°F. The tests shall be conducted as specified in 4.4.4.1.

3.9.2 Low temperature. The wiring harness, mounted in a fixture simulating installed conditions, shall withstand exposure to a temperature of -65 ± 5°F. without damage and while at this temperature shall be capable of meeting the requirements of 3.7 and 3.8. The wiring harness shall meet the requirements of 3.6 and 3.6.1 after completion of low temperature test at 80 ± 20°F. The tests shall be conducted as specified in 4.4.4.2.

3.9.3 Humidity. The wiring harness shall withstand exposure to humidity without evidence of damage affecting functioning when tested as specified in 4.4.4.3. The assembly shall meet the requirements of 3.6 through 3.8 within one hour after the conclusion of the test.

3.9.4 Transportation vibration. The wiring harness shall withstand vibration for 15 minutes in each of three mutually perpendicular axes, as shown in Figure 1, at the frequencies and amplitudes specified in Table 1 without evidence of damage or affecting function when tested as specified in 4.4.4.4. The frequency shall be varied uniformly from 8 to 500 to 8 hertz (Hz) in 15 minutes.

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After being subjected to the specified vibration, the harness shall meet the requirements of 3.6 through 3.8.

Table I

Hertz (Hz)	Cycle Time	Constant Acceleration of Double Amplitude
8-55-8	7.5 minutes	0.060 ± 0.006 inch
55-500-55	7.5 minutes	10 ± 1 g

3.9.5 Operational vibration. The wiring harness shall withstand without evidence of damage or affecting functioning a programmed sequence of band-limited white Gaussian noise and combined noise and sine wave vibration when tested as specified in 4.4.4.5. The total time of test shall be 210 seconds; 70 seconds in each of the three mutually perpendicular axes shown in Figure 1. The complex vibration test will consist of the following five sections programmed continuously:

(a) White Gaussian noise at a level of 22 g root mean square (rms) band-limited between 15 and 1500 Hertz (Hz) for one (1) second.

(b) A 4 g rms swept sine wave superimposed on 7.5 g rms of white Gaussian noise band-limited between 15 and 500 Hz for 15 seconds . The sine wave shall be swept from 800 to 1300 Hz at rate increasing directly with frequency, one such sweep taking the full 15 seconds.

(c) An 8 g rms swept sine wave superimposed on 15 g rms of white Gaussian noise band-limited between 15 and 1500 Hz for 15 seconds. The sine wave shall be swept from 1300 to 800 Hz at a rate decreasing directly with frequency, one such sweep taking the full 15 seconds.

(d) A 4 g rms sine wave superimposed on 7.5 g rms of white Gaussian noise-band-limited between 15 and 1500 Hz for 38 seconds. The sine wave shall be swept from 15 to 1500 Hz and back to 15 Hz at a rate increasing directly with frequency. One such cycle shall take the full 38 seconds.

(e) White Gaussian noise at a level of 22 g rms band-limited between 15 and 1500 Hz for one (1) second.

Following the vibration test, the wiring harness shall meet the requirements of 3.6 through 3.8.

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3.10 Soldering. All soldering shall comply with the requirements specified in Military Standard MIL-STD-1460.

3.11 Test Codes. (see 3.5). The contractor shall provide data feed back on punch cards in conformance with Quality Assurance Instruction QAI-PA-1. The data required to be coded, and the code numbers, are specified in the applicable paragraph in this section.

3.12 Workmanship.

3.12.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.12.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.12.3 Burrs. Parts shall be free from burrs.

3.12.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.12.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.12.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8830468) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

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4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table II. Tests shall be conducted in order specified in Table II, testing sequence of unit No.

Table II

First article inspection and tests

Inspection Or Test	Dest	Non Dest	Testing Sequence of Unit No. 1	2	3
1. Classification of characteristics (see 4.3.2.1)		X			
2. Continuity and resistance (see 3.6)		X	1	1	1
3. Circuit resistance (see 3.6.1)		X	2	2	2
4. Diode characteristics (see 3.7)		X	3	3	3
5. Insulation resistance (see 3.8)		X	4	4	4
6. Transportation vibration (see 3.9.4)	X		5	5	
7. High temperature (see 3.9.1)		X	7	6	7
8. Low temperature (see 3.9.2)		X		7	5
9. Operational vibration (see 3.9.5)	X				6
10. Shock (see 3.9.6)	X		6	9	
11. Humidity (see 3.9.3)	X		8	8	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision.

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4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table III and Table IV), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table III
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table IV
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Wiring Harness, Branched (prior to potting), (dwg 8830468).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Improper soldering	Visual
	102. Damaged wire insulation	Visual
	103. Loose component	Visual/Manual
	104. Sleeving missing, damaged, or loose	Visual
	105. Component missing	Visual
	106. Incorrect assembly	Visual
Minor:		
	201. Incorrect wire size	Visual
	202. Incorrect wire type	Visual
	203. Evidence of poor workmanship (see 3.13)	Visual

4.3.2.1.2 Wiring Harness, Branched (dwg 8830468).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:		
101.	Loose connector plugs	Visual/Manual
102.	Loose receptacles	Visual/Manual
*103.	Improper height of "male" terminals (Pins of the same diameter must all appear to be within 1/32 inch in height relative to the adjacent pins.)	Visual/Gage
**104.	Improper depth of "female" socket Contact spring. (Female sockets of the same diameter must all appear to be within 1/32 inch in depth relative to the adjacent sockets.)	Visual/Gage
105.	Improper cable length	Scale
106.	Insufficient potting	Visual
107.	Exposed wires	Visual
108.	Incorrect assembly	Visual
109.	Damaged component	Visual
110.	Connectors will not mate	Gage
111.	Improper mounting hole location	Gage
112.	Objective evidence relative to material missing or improper (see 3.1)	Visual
Minor:		
201.	Protective cap missing or loose	Visual/Manual
202.	Voids or air pockets greater than 1/8" diameter	Visual
203.	Incorrect or illegible marking	Visual
204.	Evidence of poor workmanship (see 3.12)	Visual
205.	Serialization improper or missing (see 3.4)	Visual

*In the event pin(s) do not appear to be within 1/32 inch in height relative to the adjacent pins, gage (see 4.3.4) shall be used to determine the variation from the normal height. Reject any pin that varies by more than 0.032 inch maximum from the normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface of all measurements.

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**In the event socket inserts do not appear to be within 1/32 inch in depth relative to the adjacent sockets, gage (see 4.3.4) shall be used to determine the variation from the normal depth. Reject any insert that varies by more than 0.032 inch maximum from the normal socket insert depth of the individual connector. The normal depth of the individual connector sockets shall be determined by calculating the average depth from measurements taken of sockets adjacent to the socket that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface for all measurements.

4.3.2.1.3 Preparation for delivery.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation and packaging not in accordance with MIL-STD-1169 (see 5.1)	Visual
102.	Packing not in accordance with MIL-STD-1169 (see 5.2)	Visual
103.	Marking not in accordance with MIL-STD-1169 (see 5.3)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.12)	Visual

4.3.3 Testing. All tests shall be performed. Tests performed in the order specified below:

	Test	Test Classification
a.	Continuity and resistance	Major
b.	Circuit resistance	Major
c.	Diode characterisitcs	Major
d.	Insulation resistance	Major
e.	Low temperature	Major
f.	Transportation vibration	Major
g.	Shock	Major

Of these tests, f and g are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

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4.3.3.1 Continuity and resistance, circuit resistance, diode characteristics, and insulation resistance. Each shall be tested in accordance with 4.4.1 through 4.4.3. If any item fails to comply with 3.6 through 3.8, the item shall be rejected and removed from the lot.

4.3.3.2 Low temperature, transportation vibration, and shock. A sample shall be selected at random in accordance with Table V. The sample shall be tested in accordance with 4.4.4.2, 4.4.4.4 and 4.4.4.6. If any item fails to comply with 3.9.2, 3.9.4 and 3.9.6, the lot shall be rejected.

Table V

Lot Size	Sample Size
2-8	1
9-180	2
181-500	3
501-800	5
801-1300	7
1301-3200	10
3201 and over	MIL-STD-105 (special inspection level S-2, AQL 0.25 percent).

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8854163). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

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4.4.1 Continuity and resistance. The continuity and resistance shall be measured between the terminals listed in 3.6 with a four-wire resistance measuring instrument. Care must be exercised to avoid bending connector terminals.

4.4.1.1 Circuit resistance. The circuit resistance between the terminals listed in 3.6.1 shall be measured with a digital ohgmeter of sufficient accuracy or a wheatstone bridge.

4.4.2 Diode characteristics. Diodes shall be tested for forward voltage drop when current specified in 3.7 is applied in the forward direction. Reverse current shall be determined by the voltage drop across a 150 ± 1.5 ohm or 100 ± 1 kilohm series resistor, as applicable, as specified in 3.7 with a voltmeter of 20,000 ohm minimum internal impedance when the voltage, as specified in 3.7, is applied in the reverse direction. The duration of the tests in either the forward or reverse direction shall be not more than 15 seconds.

4.4.3 Insulation resistance. The insulation resistance shall be measured between the terminals listed in 3.8 using a test voltage of 500 ± 50 Direct Current Volts (VDC) with a current limited to 5 Milliamperes.

4.4.4 Environmental Testing (see 3.9).

4.4.4.1 High temperature. The wiring harness shall be placed in a test chamber. The chamber's temperature shall be raised and stabilized at 160 ± 5°F. (see 6.4). After the wiring harness has been subjected to this stabilized temperature for a minimum of 4 hours and while still at this temperature, it shall be tested for compliance with the requirements of 3.7 and 3.8. After completion of the high temperature tests when the temperature of the wiring harness has returned to ambient (80 ± 20°F), the wiring harness shall be tested for compliance with the requirements of 3.6 and 3.6.1.

4.4.4.2 Low temperature. The wiring harness shall be placed in a test chamber. The chamber's temperature shall be lowered and stabilized at -65 ± 5°F (see 6.4). After the wiring harness has been subjected to this stabilized temperature for a minimum of 4 hours and while still at this temperature, it shall be tested for compliance with the requirements of 3.7 and 3.8. After completion of the low temperature testswhen the temperature of the wiring harness has returned to ambient (80 ± 20°F), the wiring harness shall be tested for compliance with the requirements of 3.6 and 3.6.1.

4.4.4.3 Humidity. The wiring harness with all of its connectors covered with potted mating connectors shall be placed in a humidity test chamber and shall be subjected to

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the humidity test as specified in MIL-E-5272, 4.4, Prodecure I, except that the number of cycles shall be three. All temperatures shall be maintained at ± 5°F. The pH value shall be adjusted by the use of chemically pure hydrochloric acid or chemically pure sodium hydroxide. Upon completion of the last cycle, the wiring harness shall be removed from the test chamber and allowed to return to ambient temperature (80 ± 20°F) without forced cooling. After returning to ambient temperature, the wiring harness may be wiped and then blown with filtered compressed air for a period of five minutes, and then shall be tested for compliance of 3.6 through 3.8.

4.4.4.4 Transportation vibration. The wiring harness shall be mounted to a fixture on a vibration machine and vibrated for 15 minutes in each of three mutually perpendicular axes, as shown in Figure I, at the frequencies and amplitudes specified in 3.9.4. Total test time for each wiring harness shall be 45 minutes. Following the vibrations, the wiring harness shall be tested for the requirements of 3.6 through 3.8.

4.4.4.5 Operational vibration. The wiring harness shall be mounted to a fixture on a vibration machine and vibrated in three mutually perpendicular axes, as shown in Figure I, using a standard noise or complex wave tape. The vibration levels shall be observed on the shake-table as near the supporting bracket as is practical. The tape shall have the characterisitcs listed in the following table.

Time (sec)	Type of Signal	Ratio of Test Level To Calibration Level	RMS g Level
0-20	Noise (calibration)(1)	1.	Set gain to give 2 g rms
20-30	None	0	0
30-31	Noise (1)	11.	22.
31-46	Noise (1) plus sinusoid (2)	4.25	8.5
	Noise alone	3.75	7.5
	Sinusoid alone	2.	4.
46-61	Noise (1) plus sinusoid (3)	8.5	17.
	Noise alone	7.5	15.
	Sinusoid alone	4.	8.
61-99	Noise (1) plus sinusoid (4)	4.25	8.5
	Noise alone	3.75	7.5
	Sinusoid alone	2.	4.
99-100	Noise (1)	11.	22.

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(1) Band limited (15 - 1500 Hz) white Gaussian Noise.
(2) Sine wave swept from 800 to 1300 Hz at a rate increasing directly with frequency. One full sweep in 15 seconds.
(3) Sine wave swept from 1300 to 800 Hz at a rate increasing directly with frequency. One full sweep in 15 seconds.
(4) Sine wave swept from 15 to 1500 Hz and back to 15 Hz at a rate increasing directly with frequency. One full cycle in 38 seconds.

For creating the random noise, the output of a General Radio random noise generator, or its equivalent, may be used if proper care is taken to insure the correct amplitude distribution of the signal. For testing the noise generator or any other associated equipment, the use of an amplitude distribution analyzer is suggested. For insuring the correct noise band-width, a filter with an asymptotic slope of at least 24 decibel (Db) per octave and 3 Db points at 15 and 1500 Hz shall be considered acceptable. Following the vibration the wiring harness shall be tested for the requirements of 3.6 through 3.8.

4.4.4.5.1 Method and procedure with the standard noise tape. Before applying the tape, the transfer function or frequency response of the entire shaking system shall be as flat as is consistent with the present state of the art. Plus or minus 3 Db or better over the band from 15 to 1500 Hz shall be considered as an objective in this respect. After proper equalization of the system, the standard noise tape is applied. The overall gain of the shake system is then adjusted so that with the noise calibration signal, the G level listed in paragraph 4.4.4.5 is obtained. With the same gain setting, the tape is run for the remainder of the test and the G levels obtained will conform to the levels listed in the table of 4.4.4.5.

4.4.4.6 Shock. The wiring harness shall be rigidly mounted to a test fixture as applicable which shall be mounted to a suitable shock producing apparatus. The wiring harness shall then be subjected to 3 shocks in each direction of the 3 major mutually perpendicular axes, as shown in Figure I. Each shock shall be of 30 $\pm$ 5 g peak amplitude and 11 $\pm$ 2 milliseconds (ms) duration as measured between points which are 10 percent of the peaks. Time to peak shall be 5.5 $\pm$ 1 ms. After completion of the 18 shocks, the wiring harness shall be examined for damage and tested for compliance with the requirements of 3.6 through 3.8.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The harness shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The harness shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The harness shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 3.4, 3.11 and 4.3.4. will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-I.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase description PA-PD-2074 (Rev. I) dated 17 September 1965.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A159

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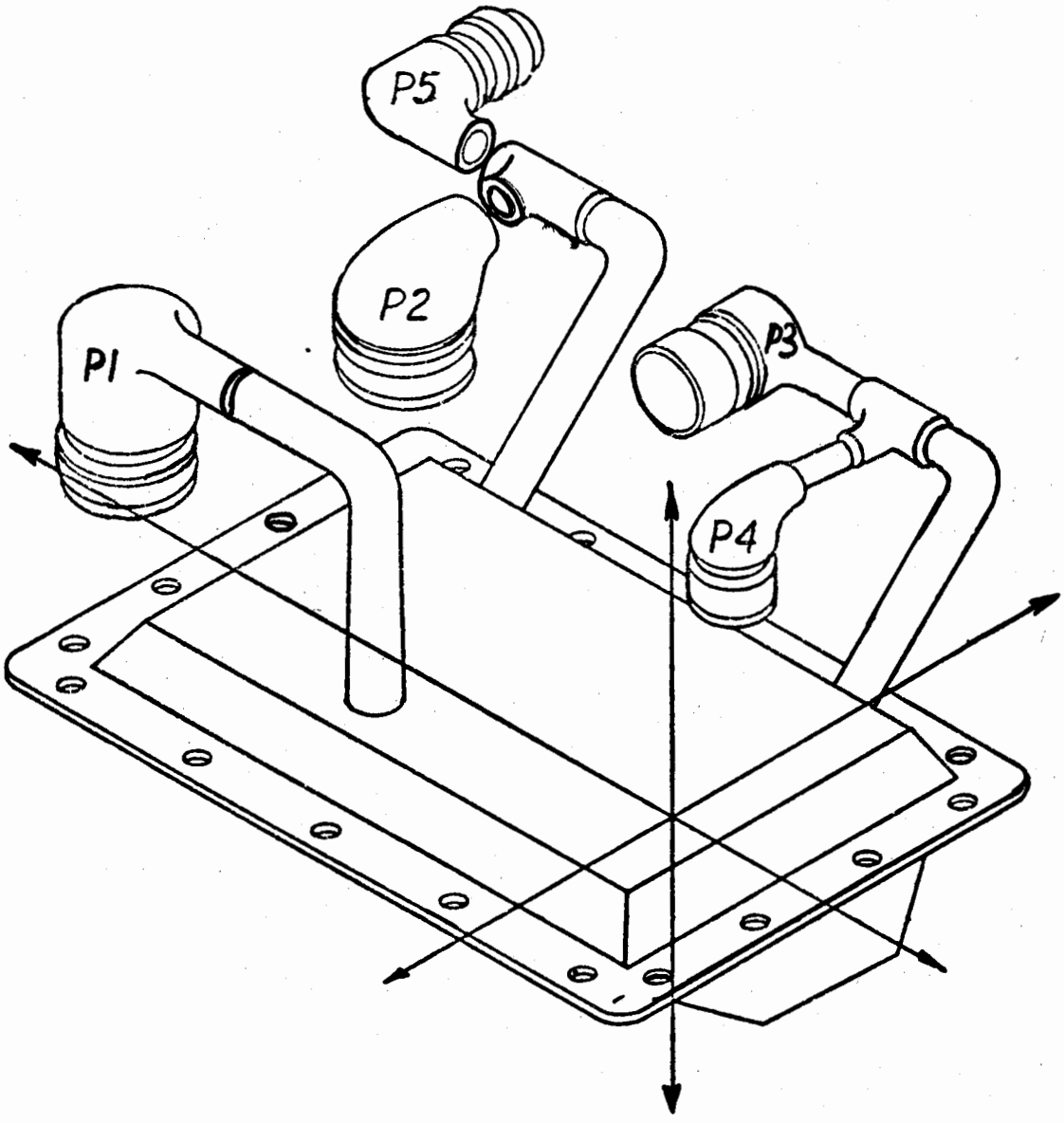


Figure 1 - Axes for vibration of the wiring harness

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